



SIMATS
ENGINEERING



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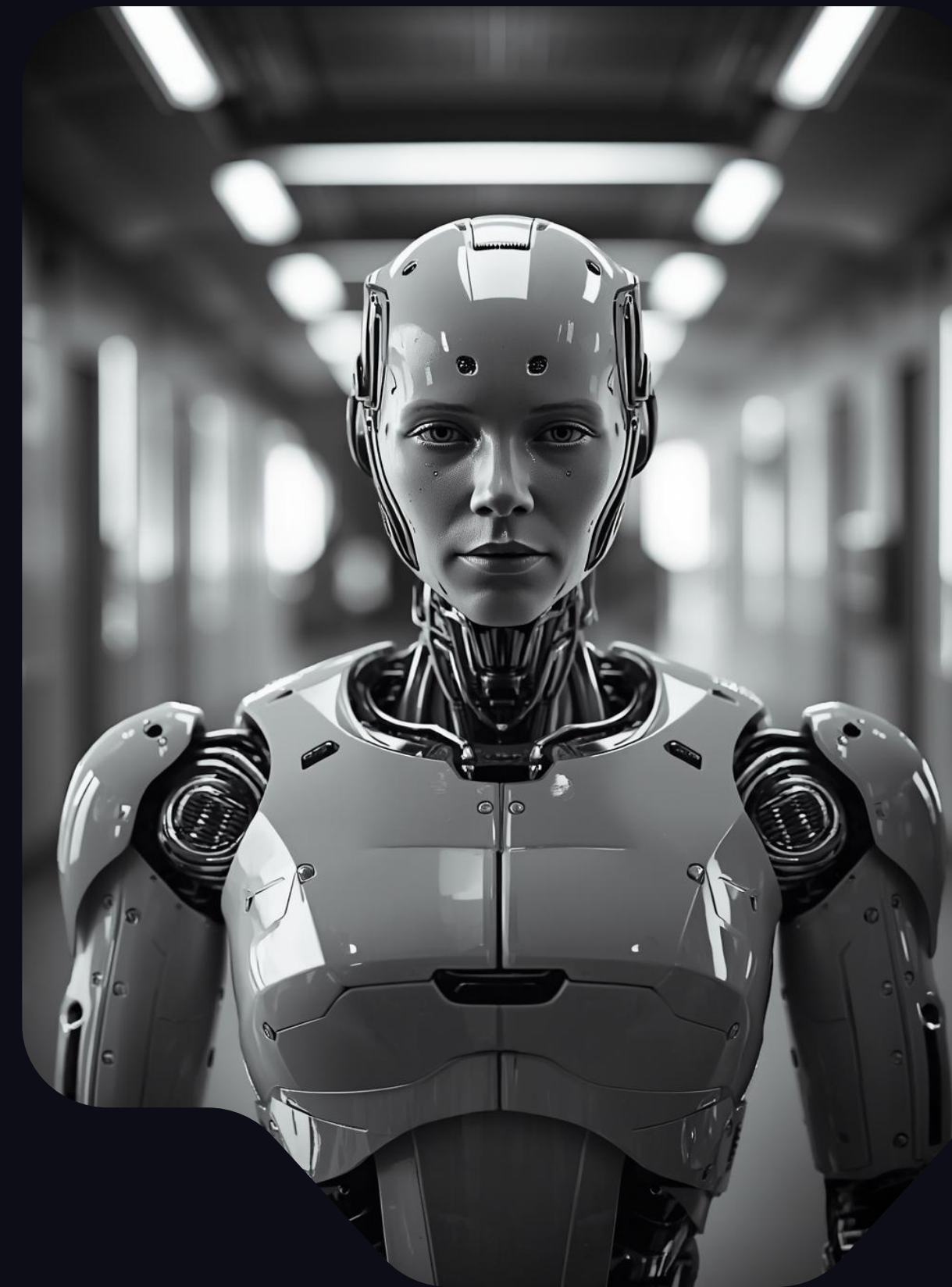
LEARNING PROBLEMS

ITA 0609 Machine Learning

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SUPERVISED,
UNSUPERVISED &
REINFORCEMENT
LEARNING

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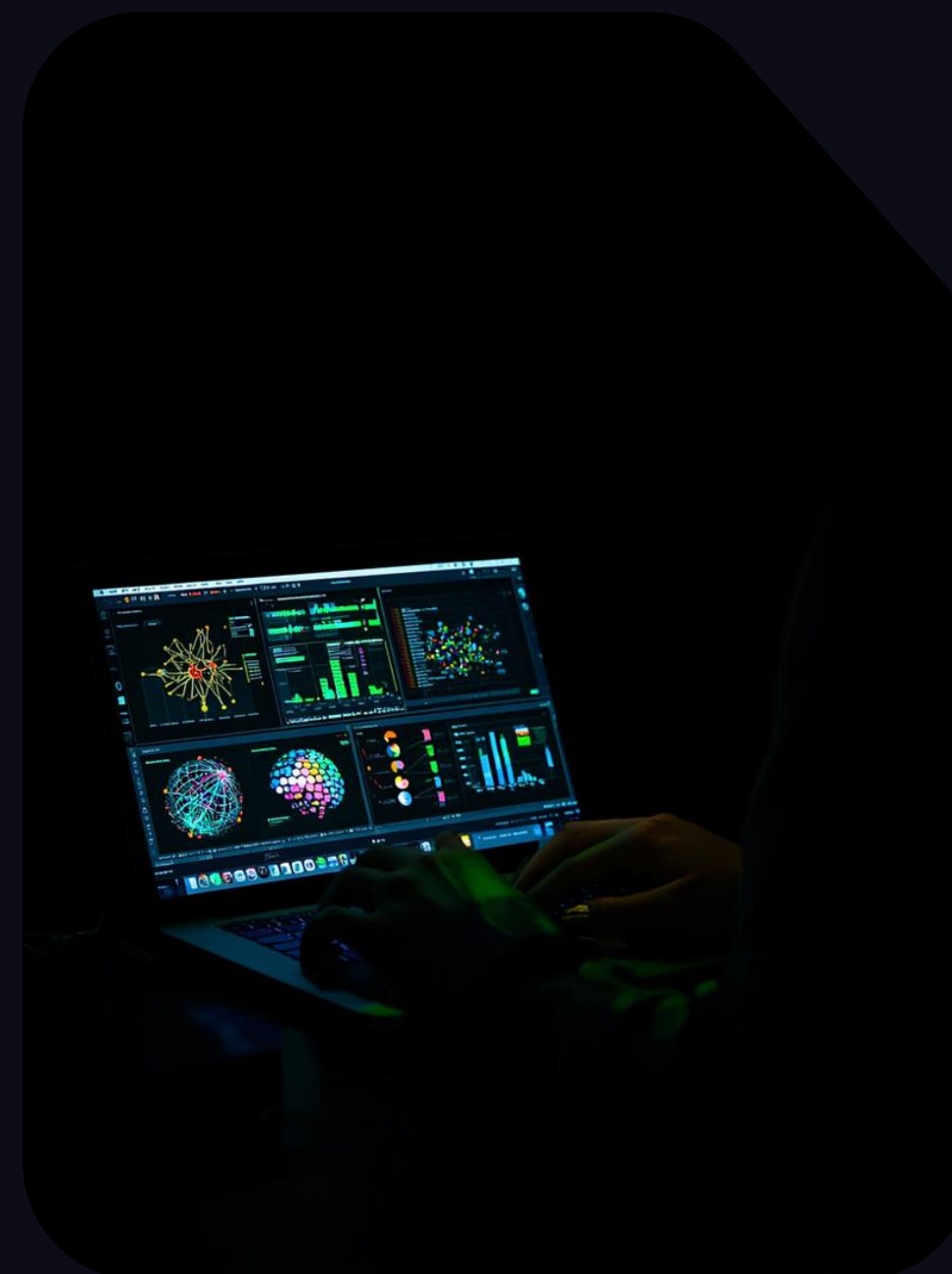




Machine learning encompasses distinct learning paradigms, each defined by how systems acquire knowledge from data and interact with their environment.

FROM SUPERVISED TO SELF-SUPERVISED

TYPES OF LEARNING PROBLEMS



TYPE 1 – SUPERVISED LEARNING

Labeled training data
Prediction & classification
Output-driven feedback

TYPE 2 – UNSUPERVISED LEARNING

No labeled examples
Pattern discovery
Clustering & grouping

TYPE 3 – REINFORCEMENT LEARNING

Agent-environment interaction
Reward-based optimization

TYPE 4 – SEMI-SUPERVISED LEARNING

Combines labeled & unlabeled data
Reduces annotation costs

TYPE 5 – SELF-SUPERVISED LEARNING

Auto-generated labels
Large-scale pretraining

LEARNING PERSPECTIVES

SYMBOLIC VS STATISTICAL APPROACHES

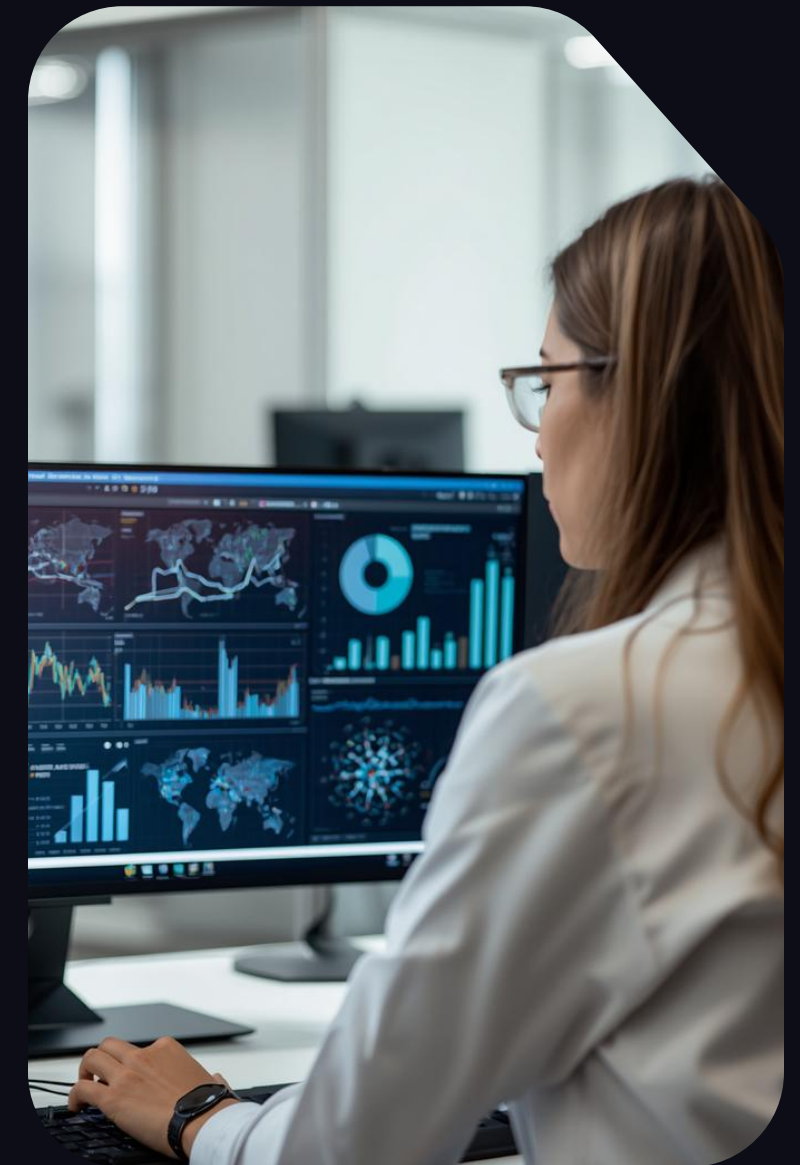
Machine learning is shaped by two core perspectives: rule-based symbolic reasoning and data-driven statistical modeling.

SYMBOLIC APPROACHES

Rule-based logic | Knowledge representation | Interpretable decisions | Expert systems

STATISTICAL APPROACHES

Data-driven models | Probabilistic inference | Neural networks | Pattern recognition





DATA QUALITY

Noise & Missing Values
Bias & Imbalance
Representativeness

OVERFITTING

Poor Generalization
Model Complexity
Regularization Needs

COMPUTATIONAL COMPLEXITY

Scalability Limits
High Training Time
Resource Constraints

MACHINE LEARNING CHALLENGES

Key issues like data quality, overfitting, and complexity directly shape the design and performance of learning systems.

KEY ISSUES IN ML

DESIGN & PERFORMANCE IMPACT

KEY INFLUENCING FACTORS

Data quality, overfitting, and computational complexity are critical factors that shape the design choices and real-world performance of machine learning systems.



IMPACT OVERVIEW

These factors determine whether a learning system generalizes well, scales efficiently, and delivers reliable predictions in production environments.

DATA QUALITY

Affects model reliability
Biased data skews outputs
Noise degrades accuracy

OVERFITTING

Reduces generalization
Poor real-world performance

COMPUTATIONAL COST

Model complexity trade-offs
Resource vs. accuracy balance





LEARNING PROBLEMS & PERSPECTIVES

CONCLUSION

Understanding learning paradigms and challenges is essential for building effective ML systems. Consider these critical factors when designing your next learning system.

KEY TAKEAWAYS

Supervised, unsupervised, and reinforcement learning address distinct problem types. Symbolic and statistical approaches offer complementary perspectives. Data quality is foundational to any effective learning system. Overfitting and generalization remain central design challenges. Computational complexity shapes algorithm selection and scalability.